EPOETIN ALFA

Target Hb ≤120 g/L

Hypertension, thrombosis

CHEMICAL NAME Recombinant human erythropoietin: 165-amino-acid glycoprotein hormone (~30.4 kDa)

CLASS Erythropoiesis-stimulating agent (ESA)

TRADE NAMES Eprex · Epogen · Procrit · Binocrit (biosimilar)

THERAPEUTIC USES / INDICATIONS

- Anaemia of chronic kidney disease (dialysis and non-dialysis).
- Chemotherapy-induced anaemia (selected patients).
- Reducing transfusion before elective orthopaedic surgery.
- Anaemia of prematurity; zidovudine-related anaemia in HIV.

ADMINISTRATION

- **SC** (preferred in non-dialysis) or **IV** (haemodialysis, into venous line at end of session).
- Store in fridge; do not shake.
- Ensure adequate iron stores (ferritin >100–200 µg/L, TSAT >20%) — give iron.

DOSAGES

- **CKD:** 50 IU/kg three times weekly (or 20,000–40,000 IU weekly for oncology); adjust by 25% based on Hb response.
- Aim for Hb 10–12 g/dL (100–120 g/L); avoid a rise >2 g/dL in 4 weeks.
- **Surgery:** 600 IU/kg weekly ×3 before surgery.

PHARMACOKINETICS

- **Onset:** reticulocytes ↑ in 7–10 days; Hb rise in 2–6 weeks.
- Bioavailability ~20% SC.
- **t½:** IV 4–5 h; SC ~24 h.

PHARMACODYNAMICS

- Binds **erythropoietin receptors** on erythroid progenitors in bone marrow → ↑ proliferation and differentiation → ↑ red-cell production.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Headache, flu-like symptoms.
- Arthralgia.
- Injection-site reactions.

ADVERSE EFFECTS (SERIOUS)

- Hypertension, hypertensive encephalopathy, seizures.
- Thrombosis: AV fistula thrombosis, stroke, MI, VTE (higher with Hb >120).
- Pure red cell aplasia (anti-EPO antibodies).
- Tumour progression (cancer patients).

CONTRAINDICATIONS & CAUTIONS

- Uncontrolled hypertension.
- Pure red cell aplasia after ESA.
- Patients unable to receive thromboprophylaxis (surgery use).
- Severe coronary/peripheral/cerebrovascular disease (surgical indication).

KEY INTERACTIONS

- **ACE inhibitors:** may ↓ response.
- **Ciclosporin:** levels may fall as haematocrit rises.

NURSING MANAGEMENT

- Monitor BP closely (especially early); hold if uncontrolled.
- Hb every 2–4 weeks during titration; iron studies.
- Check dialysis access patency.
- Teach SC self-injection.

PATIENT EDUCATION

- Have your blood pressure checked often; take BP medicines.
- Report severe headache, confusion, seizures, chest pain, swelling of legs or fistula problems.
- Keep in fridge; rotate injection sites.
- Take iron as prescribed.

KEY POINTS

- ESA for anaemia of CKD.
- Hypertension, thrombosis, seizures.
- Target Hb 10–12 g/dL.
- Needs adequate iron stores.

SOURCES CHECKED **[938]** Eprex 10000 IU/ml solution for injection in pre-filled syringe (UK SmPC) **[939]** Epogen, IV, SC (US label)

Full references p. 432–457